Po-Sheng Cheng

Master of Industrial Design, Rhode Island School of Design (RISD), Jun. 2026

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Portfolio: https://bencer3283.github.io/art/

B.Sc. Bio-Mechatronics Engineering & B.A Economics,
National Taiwan University (NTU), Jan 2023

Notable Skills

- Electronics: STM32, Arduino, ESP32, Raspberry Pi, system interface (I2C, SPI, UART), PCB design (Altium)
- Mechanical: CAD (SolidWorks, Fusion, CFD and Creo), rapid prototyping (FDM/SLA 3D Printing)
- Coding: C/C++, JavaScript (Express, React), Dart (Flutter), Python (PyTorch, NumPy), Git, Linux shell script
- Design: Rhino, Keyshot, Figma (Wireframing), Adobe Suite

Experiences

Research Assistant, RISD (directed by Prof. Peter Yeadon), Aug. 2024 - Providence, RI

- Designed and prototyped a patent-pending atmospheric water generator that can generate 500ml of water from air in 24 hours, including its mechanical structure, thermal managment and power circuit.
- Implemented a thermal and air flow simulation in Autodesk CFD to inform the design parameters of the prototype, opening up the possiblity of a architecture-scale application.

Apprentice, Google Hardware Product Sprint, Jun. - Sep. 2024 Taipei, Taiwan

- Led a team of 5 to design a Hardware as a Service platform for managing lost items in public facilities. Laid out the system architecture for both the hardware kiosk and the web services.
- My prototypes of a microcontroller-based hardware, a mobile app and a RESTful API service drove the development direction from a single hardware product to a integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022 Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating a microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype recieved widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of the development.

(Publication) TeleSHift: Telexisting TUI for Physical Collaboration & Interaction

This work recieved the **Best Demo Award at Ubicomp/ISWC 2022 Conference**, https://doi.org/10.1145/3544793.3560323

- Utilized microcontroller, potentiometers, DC motors and touch sensors to implement a shape-transforming device with a 3D tangible user interface that enabled this work to be awarded at ACM Ubicomp conference.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.

College Student Researcher, NTU, Jul. 2021 - Feb. 2022 Taipei, Taiwan

- Developed a novel, integrated spectral mapping system that vastly accelerates scientific spectral measurement by being involved in research projects to understand the bottleneck of existing workflow.
- By using linear dispersion optics and optimizing the EMCCD readout algorithm in LabVIEW and C++, the scanning time was reduced by 26%.
- Awarded the **2021 Technology Innovation Award** by CCMS, NTU and **College Student Research Creativity Award** by National Science and Technology Council of Taiwan with this project.

Project Lead, Bio-Electromagnetics Laboratory, NTU, May. 2020 - Jul. 2022 Taipei, Taiwan

- Designed an device to monitor the amount of insects in farm fields including a custom PCB with wireless capabilities and rigorous mechanisms.